FINAL PROJECT REVIEW

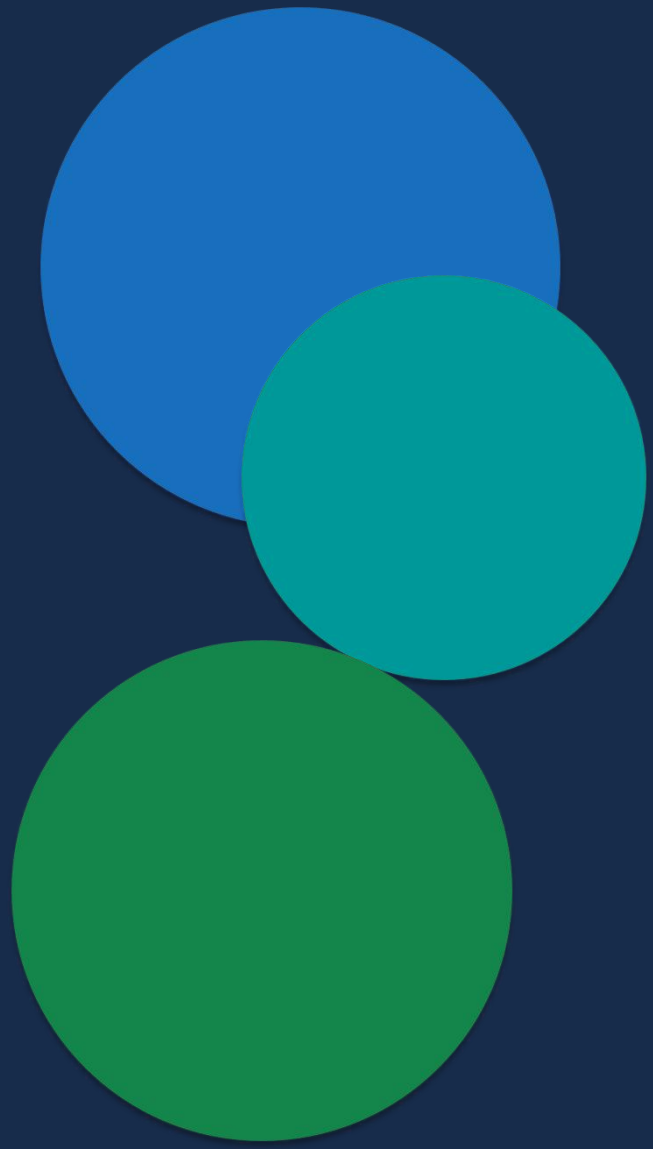
Circular E-Waste Recovery & Refurbishment Intelligence Platform

A connected two-application workflow for collection, recovery, device assessment and refurbishment coordination

PROJECT TEAM

Application 1: Collection & Recovery – Nideesh C G (Reg.No: 23MIC0092)

Application 2: Refurbishment Intelligence –Dharma S (Reg.No: 23MIC0104)



FINAL REVIEW • 2026

Project Overview

Two independent applications connected through a shared REST API.

01 Collection & Recovery

- Capture electronic-device collection records
- Maintain device details and current status
- Manage images and supporting documents
- Prepare eligible devices for refurbishment

02 Refurbishment Intelligence

- Receive device information from Application 1
- Assess device condition and technical notes
- Record refurbishment decisions and outcomes
- Return results through the API

Shared integration layer

- REST API-based communication
- Structured JSON data exchange
- Authenticated requests
- Bidirectional status and result updates

Core idea **Connect separate workflows while keeping each application independently maintainable.**

Problem Statement

E-waste workflows span multiple operational stages and need consistent information flow.

Fragmented tracking

- Collection information may be stored separately
- Device status can be difficult to follow
- Repeated manual entry can cause inconsistencies

Limited coordination

- Collection and refurbishment teams need shared records
- Assessment outcomes must reach the originating workflow
- Documents and metadata need traceability

Connected workflow

- Separate applications with clear responsibilities
- Secure and structured API communication
- A clearer path from collection to recovery decisions

Project focus: improve coordination and traceability through a connected software workflow.

Project Objectives

Operational tracking combined with cross-application data exchange.

Digitize collection

Capture and manage device-level collection information.

Enable assessment

Share eligible device details with the refurbishment workflow.

Return outcomes

Transfer assessment status and agreed result fields.

Support reporting

Provide dashboards, summaries and reports where implemented.

Keep modules independent

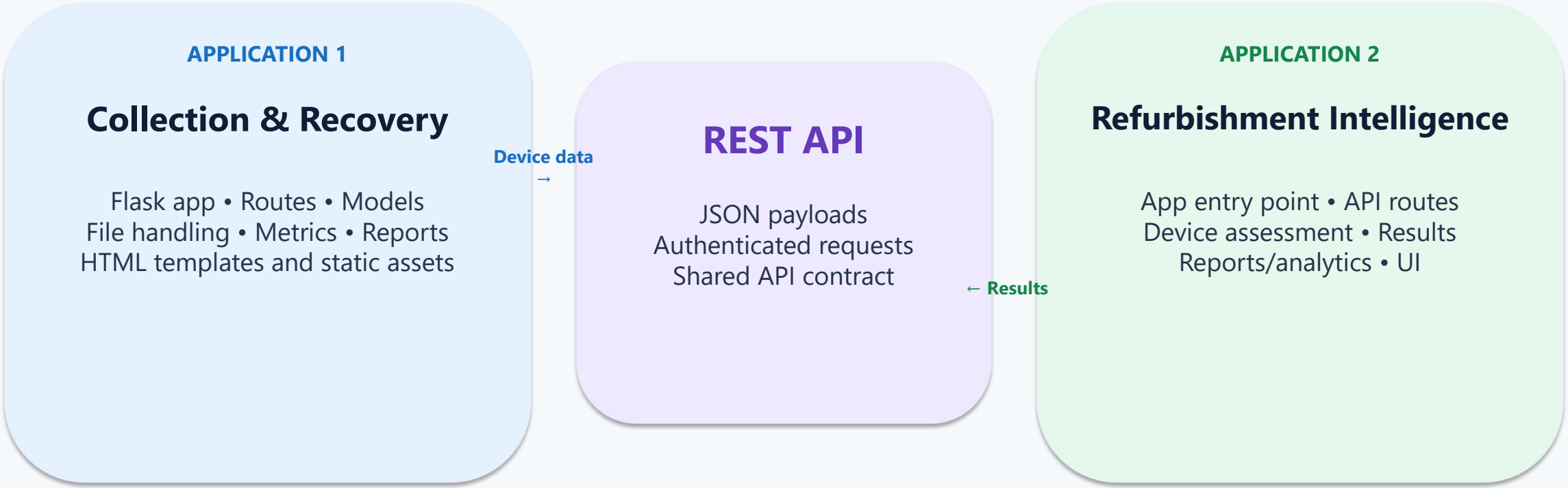
Use a common REST API contract between applications.

Prepare for hosting

Configure production settings and PostgreSQL data persistence.

System Architecture

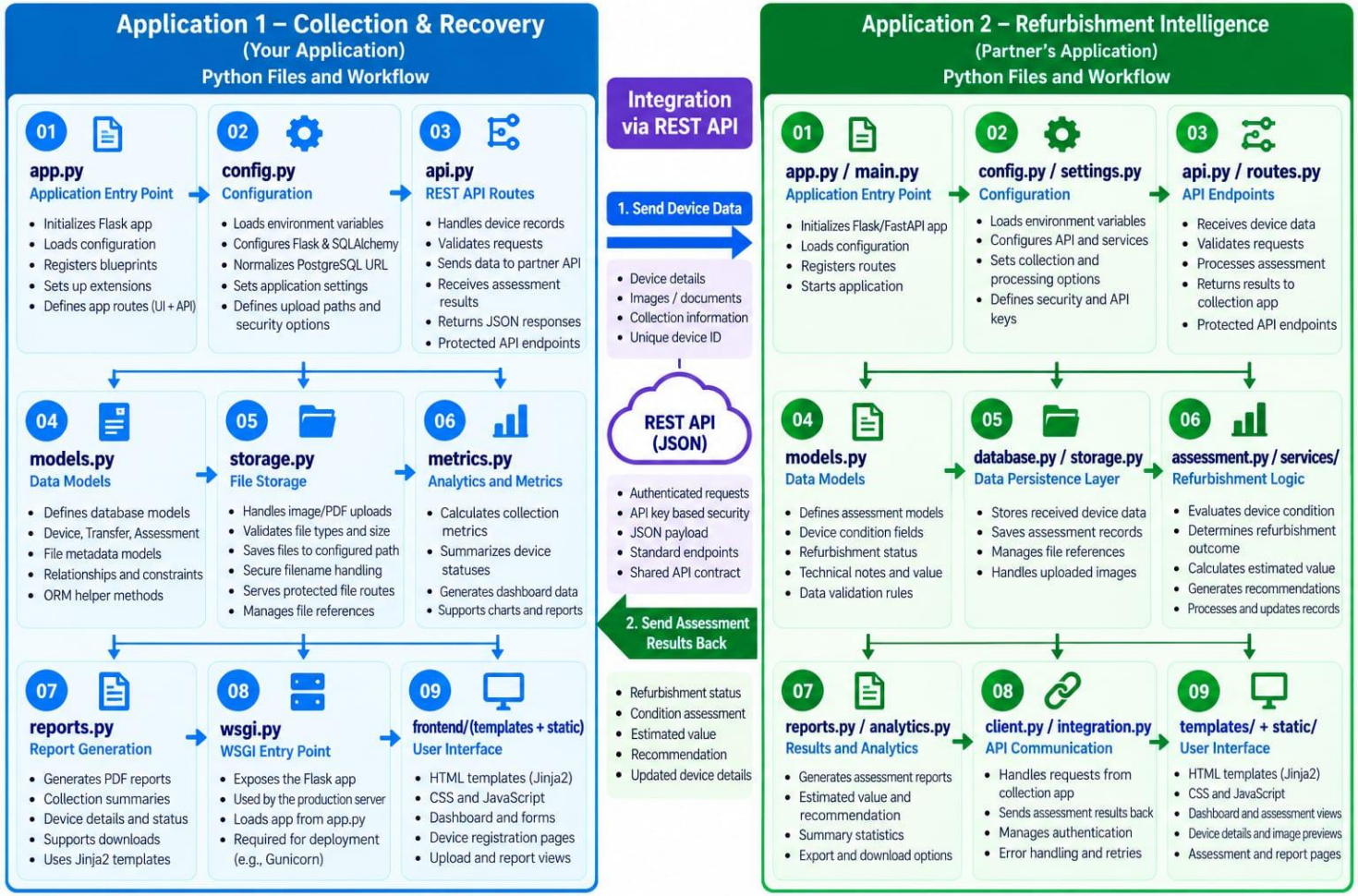
High-level view of the two applications and their integration boundary.



Each application manages its own codebase; the REST API defines how information is exchanged.

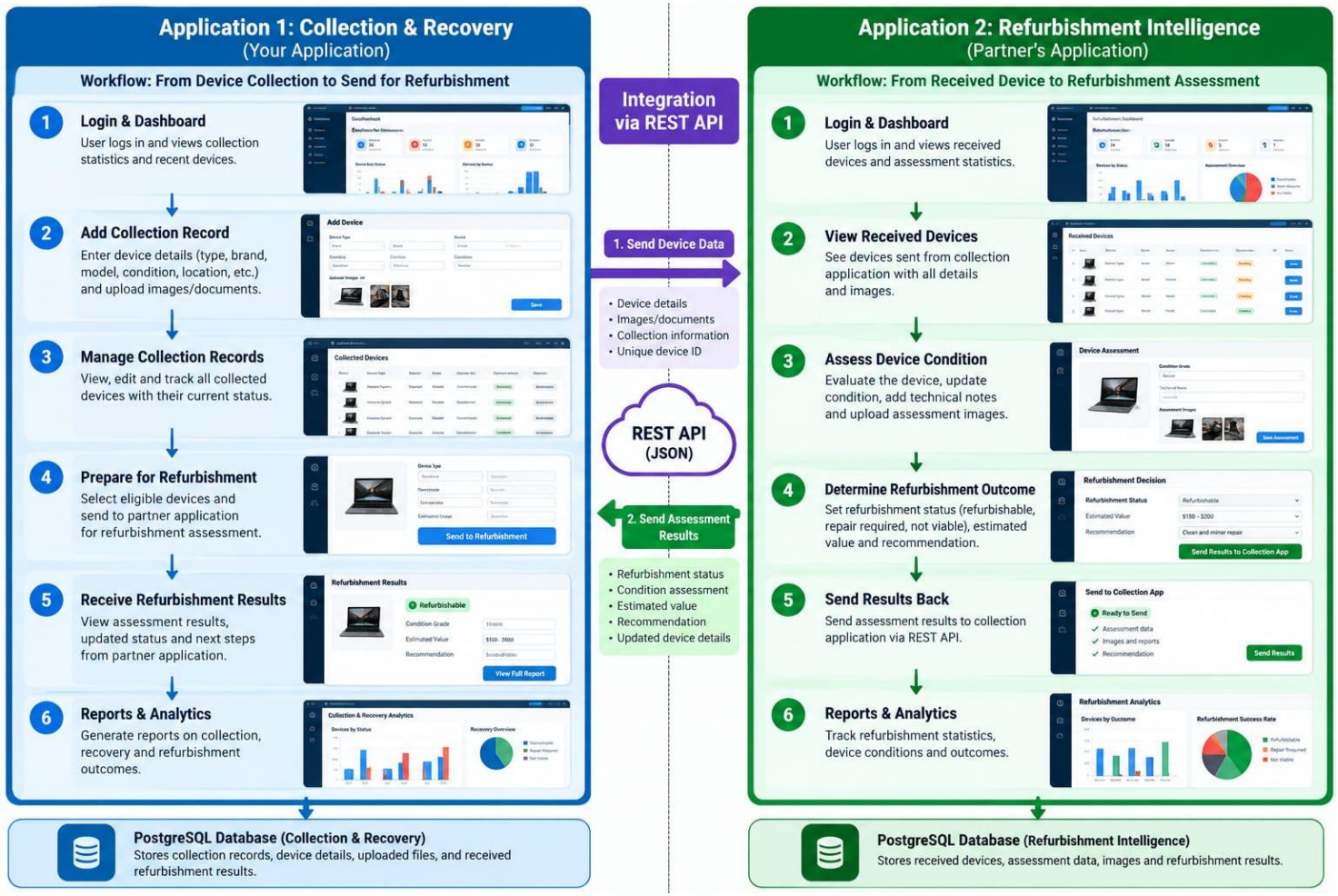
Python Files Workflow — Both Applications

Module-level view of Python files, responsibilities and API integration.



End-to-End Application Workflow

From device collection through refurbishment assessment and results return.



Application 1 — Collection & Recovery

Primary responsibilities of the collection-side application.

Capture and manage

- Register device and collection details
- Maintain status and metadata
- Support image/document workflows where enabled

Process and present

- Expose UI and API routes
- Calculate collection metrics and summaries
- Present dashboards, records and report views

Coordinate with partner

- Select devices for refurbishment
- Send details through the REST API
- Receive and display returned assessment results

Representative files: app.py, api.py, config.py, models.py, storage.py, metrics.py, reports.py, wsgi.py, migrations/, templates and static assets.

Application 2 — Refurbishment Intelligence

Partner-side responsibilities; confirm exact filenames against the partner repository.

Receive device data

- Expose endpoints for incoming records
- Validate received device information
- Link images/documents where supported

Assess and decide

- Review device condition and technical notes
- Record refurbishment status and outcomes
- Provide estimates or recommendations if implemented

Return results

- Send assessment data to Application 1
- Handle API errors and authentication
- Support assessment summaries and reports

Representative roles: entry point, configuration, API routes, data models, persistence, assessment logic, integration client, reports and UI.

Bidirectional API Data Exchange

The REST API is the communication boundary between independently developed applications.

01 **Collection → Refurbishment**

Device identifier, collection details, condition metadata and supported file references.

02 **Refurbishment → Collection**

Assessment status, outcome, notes, recommendations and agreed result fields.

Implementation checks: shared API contract • authentication • validation • error handling • duplicate-request safety

Technology Stack & Deployment

Independent services, production configuration and API-based connection.

Application layer

- Python backend
- Flask for the collection application
- HTML templates, CSS and JavaScript UI
- Confirm partner framework from its repository

Integration & data

- REST API with JSON payloads
- Environment-based configuration
- PostgreSQL planned for production
- SQLite may be retained for local development

Hosting readiness

- Deploy each app independently
- Set production secrets securely
- Apply migrations safely
- Test health endpoints and file persistence

State hosting as completed only after the live deployment and end-to-end tests succeed.

Validation & Testing

Demonstrate the complete workflow, not just individual screens.

UI	Login, dashboard, records and navigation
DATA	Create and retrieve device records
API	Valid request, invalid credential and error response
INTEGRATION	Send a device record and receive assessment results
FILES	Upload, preview, download and persistence checks
DATABASE	Verify records survive redeploy/restart

Only mark a test as passed after demonstrating it in the live or local application.

PROJECT OUTCOME

A connected workflow for a more circular electronics lifecycle

Collection & recovery → Device assessment → Refurbishment outcome → Traceable records

Traceability

Keep device information connected across stages.

Coordination

Exchange data between separate applications.

Reuse potential

Support informed recovery decisions.

Thank you

Questions & Discussion